US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12414263
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Yang; Che-Yuan

Cooling tray connector assembly

Abstract

An example connector assembly includes a cage, a first liquid cooling tray, a second liquid cooling tray, and a pressuring spring. The cage includes a frame and partitioning walls. The frame and the partitioning walls define an insertion space. The first liquid cooling tray is provided to a top of the cage, with a lower surface of the first liquid cooling tray constituting an upper wall surface of the insertion space. The second liquid cooling tray is provided to a bottom of the cage, with an upper surface of the second liquid cooling tray constituting a lower wall surface of the insertion space. The pressuring spring is positioned within the insertion space.

Inventors: **Yang; Che-Yuan (New Taipei, TW)**

Applicant: **Molex, LLC** (Lisle, IL)

Family ID: **78911814**

Assignee: **Molex, LLC (Lisle, IL)**

Appl. No.: **18/211307**

Filed: **June 19, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230337393 A1	Oct. 19, 2023

Foreign Application Priority Data

CN	202010559236.4	Jun. 18, 2020

Related U.S. Application Data

continuation parent-doc US 17349976 20210617 US 11729941 child-doc US 18211307

Publication Classification

Int. Cl.: **H01R4/60** (20060101); **H01R13/518** (20060101); **H05K7/20** (20060101)

U.S. Cl.:

CPC **H05K7/20218** (20130101); **H01R13/518** (20130101);

Field of Classification Search

CPC: H01R (13/518); H05K (7/20218)

USPC: 439/199

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4644385	12/1986	Nakanishi	257/E23.091	H01L 23/473
5473508	12/1994	Porter	257/713	H01L 23/467
6816376	12/2003	Bright	165/185	G02B 6/4201
6980437	12/2004	Bright	361/704	H04B 1/036
6986679	12/2005	Aronson et al.	N/A	N/A
7371965	12/2007	Ice	N/A	N/A
7539018	12/2008	Murr	439/372	H05K 7/20418
7625223	12/2008	Fogg	N/A	N/A
8223498	12/2011	Lima	N/A	N/A
8345426	12/2012	Nichols	361/756	H05K 7/1418
8382509	12/2012	David	439/342	H01R 13/6471
8432694	12/2012	Hentschel	165/80.4	H05K 7/20218
8444437	12/2012	Szczesny	439/607.17	H01R 13/6587
8449331	12/2012	Phillips	439/607.21	H01R 13/6581
8556658	12/2012	Szczesny	439/607.2	G02B 6/4278
8670236	12/2013	Szczesny	174/358	H05K 9/0009
8879262	12/2013	Shi	165/185	G02B 6/4246
8879267	12/2013	Henry	361/709	H04L 12/6418
9258925	12/2015	Arvelo	N/A	G06F 1/20
9793648	12/2016	Regnier	N/A	G02B 6/4201
9913403	12/2017	Krug, Jr.	N/A	H05K 7/20781
9935403	12/2017	Briant	N/A	H05K 9/0018
9960525	12/2017	Regnier	N/A	G02B 6/4201

9980411	12/2017	Yang	N/A	N/A
10104760	12/2017	Briant	N/A	G02B 6/4284
10249983	12/2018	Regnier et al.	N/A	N/A
10276995	12/2018	Little	N/A	G02B 6/4284
10278313	12/2018	Jia	N/A	H05K 1/144
10321607	12/2018	Ito	N/A	G02B 6/42
10539753	12/2019	Leigh	N/A	N/A
10709034	12/2019	Jochim	N/A	H05K 7/20781
10765038	12/2019	Leigh	N/A	H05K 7/20772
10795096	12/2019	Leigh	N/A	G02B 6/4448
10795100	12/2019	Leigh	N/A	G06F 1/185
10856446	12/2019	Boyden	N/A	G06F 1/183
11165185	12/2020	Lloyd	N/A	H01R 13/502
11483943	12/2021	Leigh	N/A	H05K 7/20772
2005/0168932	12/2004	Selvidge	361/679.02	H05K 7/1425
2009/0109627	12/2008	Murr	361/704	H05K 7/20418
2013/0157499	12/2012	Crippen et al.	N/A	N/A
2015/0077937	12/2014	Daly	361/699	G02B 6/4268
2016/0290728	12/2015	Coteus	N/A	H05K 7/20254
2019/0246523	12/2018	Boyden et al.	N/A	N/A
2021/0153385	12/2020	Lu	N/A	H05K 7/20254

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102067063	12/2010	CN	N/A
103201911	12/2015	CN	N/A
105445859	12/2016	CN	N/A
106304754	12/2018	CN	N/A
110058358	12/2018	CN	N/A
108061943	12/2019	CN	N/A
110658595	12/2019	CN	N/A
110806620	12/2019	CN	N/A
111083903	12/2019	CN	N/A
212160165	12/2019	CN	N/A
111033916	12/2020	CN	N/A
114966998	12/2021	CN	N/A
115275716	12/2021	CN	N/A
112993659	12/2022	CN	N/A
111208611	12/2023	CN	N/A

OTHER PUBLICATIONS

Notice of Allowance received for U.S. Appl. No. 17/349,976, mailed on Mar. 21, 2023, 12 pages. cited by applicant

Primary Examiner: Patel; Harshad C

Assistant Examiner: Imas; Vladimir

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/349,976, filed Jun. 17, 2021, titled “CONNECTOR ASSEMBLY,” and claims the benefit of priority to Chinese Patent Application No. 202010559236.4, filed Jun. 18, 2020, the entire contents of both of which applications are hereby incorporated herein by reference.

TECHNICAL FIELD
(1) The present disclosure relates to a connector assembly, and particularly relates to a connector assembly having a liquid cooling tray.
BACKGROUND
(2) U.S. Pat. No. 7,625,223 B1 discloses a receptacle assembly which includes a guide frame, a heat sink, and a conductive gasket. The guide frame includes a top wall and defines an internal cavity configured to receive a mating connector. The top wall defines an opening which is communicated with the internal chamber. The heat sink is held in the internal cavity such that an upper portion of the heat sink passes through the opening and a lower portion of the heat sink is engaged with the mating connector when the mating connector is inserted into the internal cavity. The conductive gasket is held between the guide frame and the heat sink. The heat sink includes an engagement portion, when the mating connector is inserted into the internal cavity, the engagement portion provides a thermally conductive pathway for the mating connector. The conductive gasket is configured to be compressed between the guide frame and the heat sink so that the conductive gasket provides a conductive path between the heat sink and the guide frame. However, the heat sink disclosed in this prior art is generally a metal heat sink with heat dissipation fins, a principle thereof is that heat is brought out by an air flow between the heat dissipation fins, heat dissipation performance of the heat sink is slightly insufficient when the heat sink is used in a connector assembly which has a higher transmission speed and generates more heat.
(3) Chinese invention patent application publication No. CN110139534A (corresponding to United States patent application publication No. US2019/0246523A1) discloses a cooling device which includes a manifold and a plurality of pedestals. The manifold includes a housing that encloses an interior cavity, the interior cavity is used to receive a cooling fluid and circulate the cooling fluid. Each of the pedestals is individually and flexibly coupled to the housing of the manifold via a bellows which is used to seal and is ring-shaped. Each pedestal is configured to extend outwardly from a bottom surface of the housing when a fluid pressure is present within the interior cavity. However, the bellows in this prior art only provides a flexible connection between the pedestal and the manifold, only there is a fluid pressure in the inner cavity of the manifold can the pedestal have a sufficient energy to extend outwardly from the bottom surface of the housing of the manifold. Firstly, the bellows as a connection sealing member between the pedestal and the manifold is not only complex in structure but also difficult to manufacture; secondly, because only the pedestal needs to use a fluid pressure can the pedestal extend outwardly, when the fluid pressure is insufficient or unstable, it is prone to cause a protruding amount of the pedestal which protrudes from the manifold insufficient, which results in a pressure of the pedestal contacting an electrical module is insufficient or even the pedestal cannot contact the electrical module, in turn heat dissipation efficiency is lowered. In addition, when a plurality of electrical modules need to contact the individual pedestals on the manifold at the same time, with respect to a case that the protruding

amount of each pedestal which protrudes from the manifold is insufficient and an elastic restoring force of each pedestal relative to the manifold is insufficient, due to tolerances of individual electrical modules, the individual electrical modules cannot contact the respective pedestals on the manifold at the same time.

SUMMARY

(4) Therefore, an object of the present disclosure is to provide a connector assembly which can improve at least one of the deficiencies in the prior art.

(5) An example connector assembly includes a cage, a first liquid cooling tray, a second liquid cooling tray, and a pressuring spring. The cage includes a frame and partitioning walls. The frame and the partitioning walls define an insertion space. The first liquid cooling tray is provided to a top of the cage, with a lower surface of the first liquid cooling tray constituting an upper wall surface of the insertion space. The second liquid cooling tray is provided to a bottom of the cage, with an upper surface of the second liquid cooling tray constituting a lower wall surface of the insertion space. The pressuring spring is positioned within the insertion space.

(6) In other aspects, the pressuring spring is provided on the upper surface of the second liquid cooling tray within the insertion space. The pressuring spring can include a plurality of spring plates extending into the insertion space from the upper surface of the second liquid cooling tray in one example. In another example, the pressuring spring includes a mounting portion assembled to the upper surface of the second liquid cooling tray and a spring plate extending from the mounting portion into the insertion space.

(7) In other aspects, the mounting portion of the pressuring spring can include a plate body assembled to the upper surface of the second liquid cooling tray and a frame body folded back from the plate body onto an upper surface of the plate body, and the spring plate can extend from an inner edge of the frame body into the insertion space. In another example, the mounting portion of the pressuring spring can be positioned at a front end of the pressuring spring, and the spring plate can extends from the mounting portion into the insertion space.

(8) In other aspects, the partitioning walls can be assembled to the first liquid cooling tray and the second liquid cooling tray. In one example, a first positioning recess can be formed on the lower surface of the first liquid cooling tray, a second positioning recess can be formed on the upper surface of the second liquid cooling tray, and a partitioning wall among the partitioning walls includes positioning protrusions which are snapped into the first positioning recess and the second positioning recess.

(9) In other examples, the partitioning walls can define a number of insertion spaces, the lower surface of the first liquid cooling tray can constitute an upper wall surface of the plurality of insertion spaces, and the upper surface of the second liquid cooling tray can constitute a lower wall surface of the plurality of insertion spaces. The insertion space can include a first insertion space among a number of insertion spaces, and the connector assembly further includes a second pressuring spring within a second insertion space among the number of insertion spaces.

(10) Another example connector assembly includes a cage, a first liquid cooling tray, a second liquid cooling tray, a first pressuring spring, and a second pressuring spring. The cage includes an upper cage, a lower cage, a frame, and partitioning walls. The frame and the partitioning walls define an upper insertion space in the upper cage and a lower insertion space in the lower cage. The first liquid cooling tray is provided to a top of the upper cage, with a lower surface of the first liquid cooling tray constituting an upper wall surface of the upper insertion space. The second liquid cooling tray is provided between the upper cage and the lower cage, with an upper surface of the second liquid cooling tray constituting a lower wall surface of the upper insertion space and a lower surface of the second liquid cooling tray constituting an upper wall surface of the lower insertion space. The first pressuring spring is positioned within the upper insertion space, and the second pressuring spring is positioned within the lower insertion space.

(11) Another example connector assembly includes a cage, a liquid cooling tray, and a pressuring

spring. The cage includes a frame and partitioning walls. The frame and the partitioning walls define an insertion space. The liquid cooling tray is provided to a top of the cage, with a lower surface of the liquid cooling tray constituting an upper wall surface of the insertion space. The pressuring spring is positioned within the insertion space.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other features and technical effects of the present disclosure will be apparent in an embodiment referring to the accompanying figures, in which:

(2) FIG. **1** is a perspective exploded view of a first embodiment of a connector assembly of the present disclosure, a circuit board and a pluggable module;

(3) FIG. **2** is a perspective exploded view of the first embodiment;

(4) FIG. **3** is a perspective exploded view of a cage of the first embodiment positioned in the upper;

(5) FIG. **4** is a perspective exploded view of a cage of the first embodiment positioned in the lower;

(6) FIG. **5** is a perspective exploded view of the cages, a first liquid cooling tray and a second liquid cooling tray of the first embodiment;

(7) FIG. **6** is a perspective exploded view of the cages, the first liquid cooling tray and the second liquid cooling tray of the first embodiment of FIG. **5** viewed from another angle;

(8) FIG. **7** is a perspective exploded view of the second liquid cooling tray and a pressuring spring of the first embodiment;

(9) FIG. **8** is a cross sectional view of the first embodiment;

(10) FIG. **9** is a perspective exploded view of a second embodiment of the connector assembly of the present disclosure;

(11) FIG. **10** is a perspective exploded view of a cage of the second embodiment positioned in the upper;

(12) FIG. **11** is a perspective exploded view of a cage of the second embodiment positioned in the lower;

(13) FIG. **12** is a perspective exploded view of the cages, a first liquid cooling tray and a second liquid cooling tray of the second embodiment; and

(14) FIG. **13** is a perspective exploded view of the cages, the first liquid cooling tray and the second liquid cooling tray of the second embodiment of FIG. **12** viewed from another angle.

DETAILED DESCRIPTION

(15) Before the present disclosure is described in detail, it is noted that like elements are represented by the same reference numerals in the following description.

(16) Referring to FIG. **1** and FIG. **2**, a first embodiment of a connector assembly **100** of the present disclosure is adapted to be provided to a circuit board **200** and is adapted to be mated with a pluggable module **300**. The connector assembly **100** includes two cages **1**, a plurality of receptacle connectors **2**, a first liquid cooling tray **3**, a second liquid cooling tray **4** and a plurality of pressuring springs **5**. It is noted that, for sake of clearly illustration and easy understanding, only one of the plurality of receptacle connectors **2** and one of the pluggable modules **300** are shown in FIG. **1**.

(17) Referring to FIGS. **1** to **4**, the two cages **1** are arranged in an up-down direction D**1** (a direction pointed by an arrow of D**1** is up, and a direction opposite thereto is down). Hereinafter, a cage positioned in the upper is referred to as an upper cage **1***a*, and a cage positioned in the lower is referred to as a lower cage **1***b*. Each cage **1** includes a frame **11** and a plurality of partitioning walls **12** which are provided to the frame **11**, are arranged side by side along a left-right direction D**2** (a direction pointed by an arrow of D**2** is right, and a direction opposite thereto is left) and are spaced apart from each other. In the first embodiment, the frame **11** is formed by metal die-casting, the

plurality of partitioning walls **12** are formed for example by punching a metal sheet, so the frame **11** as a main structure of the cage **1** has a higher strength and more manufacturing cost is saved for the plurality of partitioning walls **12**, requirements of high strength and low cost are considered at the same time. However, in other embodiments, the frame **11** and the plurality of partitioning walls **12** may be formed integrally or separately in other manufacturing manners. The plurality of partitioning walls **12** are assembled to the frame **11** in form of assembling configuration. The frame **11** and the plurality of partitioning walls **12** together define a plurality of insertion spaces **13** which are arranged transversely along the left-right direction D**2** and each are opened forwardly in a front-rear direction D**3** (a direction pointed by an arrow of D**3** is front and a direction opposite thereto is rear), the plurality of insertion spaces **13** each allow the pluggable module **300** to insert therein In the first embodiment, the lower cage **1***b* further includes a bottom plate **15** which is assembled to the frame **11** and constitutes a lower wall surface of each of the plurality of insertion spaces **13** of the lower cage **1***b*. The bottom plate **15** is formed by a metal sheet in the first embodiment.

(18) Specifically, the frame **11** of each cage **1** has a front frame portion **111** which is positioned at a front end of the frame **11** and extends along the left-right direction D**2**, two side frame portions **112** which extend rearwardly from a left side and a right side of the front frame portion **111** respectively, and a rear frame portion **113** which is connected between the two side frame portions **112** and is positioned behind the front frame portion **111**. A front end of the front frame portion **111** is formed with an insertion frame **111***a* which extends forwardly and surrounds a front end of the plurality of insertion spaces **13**. Each partitioning wall **12** is formed with a plurality of engaging pieces **121** which bend laterally from front and rear ends of a top edge and a bottom edge of each partitioning wall **12**. The insertion frame **111***a* and the rear frame portion **113** of the frame **11** are formed with a plurality of engaging grooves **114** which extend from the front to the rear and allow the plurality of partitioning walls **12** respectively to insert rearwardly and allow the plurality of engaging pieces **121** to correspondingly engage with the insertion frame **111***a* and the rear frame portion **113** of the frame **11** respectively. Each side frame portion **112** of each cage **1** is formed with a positioning groove **112***a* which extends upwardly from a bottom of each side frame portion **112** and a latching block **112***b* which protrudes outwardly from an outer side surface of each side frame portion **112**. The upper cage **1***a* further includes two locking members **14** which are assembled to the two side frame portions **112** of the frame **11** respectively. Each locking member **14** includes a locking portion **140** which is positioned at an inner side surface of the side frame portion **112** and a fixing portion **142** which is latched to the outer side surface of the corresponding side frame portion **112**. The locking member **140** has a locking piece L thereon, the locking portion **140** is provided with a positioning piece **141** which is snapped into the positioning groove **112***a* of the corresponding side frame portion **112**. Furthermore, the positioning piece **141** further can hook the outer side surface of the side frame portion **112**, the fixing portion **142** is formed with a latching hole **142***a* which correspondingly receives the latching block **112***b* The bottom plate **15** of the lower cage **1***b* has a bottom plate portion **151**, locking portions **152** which extend upwardly respectively from a left side edge and a right side edge of the bottom plate portion **151** respectively, and latching portions **153** which extend upwardly from the left side edge and the right side edge of the bottom plate portion **151** respectively and each are positioned at an outside of the corresponding locking portion **152**. The locking portion **152** is assembled to an inner side surface of the corresponding side frame portion **112**, the latching portion **153** is latched to the outer side surface of the corresponding side frame portion **112**. The locking portion **152** is formed with a positioning piece **152***a* which is snapped into the positioning groove **112***a* of the corresponding side frame portion **112**. Furthermore, the positioning piece **152***a* further can hook the outer side surface of the side frame portion **112**, the latching portion **153** is formed with a latching hole **153***a* which correspondingly receives the latching block **112***b*. In addition, it is noted that, the partitioning wall **12**, the locking portion **140** of the locking member **14** and the locking portion **152** each are formed with a locking piece L which protrudes into the insertion space **13**. When the pluggable module **300**

is inserted into the insertion space **13**, the locking member L snaps into a locking groove **301***a* which is formed to each of both sides of a mating portion **301** of the pluggable module **300**, so that the pluggable module **300** is locked in the insertion space **13**. In addition, each cage **1** further includes a plurality of first elastic grounding elements **16** which are assembled to the insertion frame **111***a* of the frame **11** and a plurality of second elastic grounding members **17** which are assembled at front ends of the plurality of partitioning walls **12**. Each first elastic grounding member **16** has a first inner side elastic finger **161** which is provided at an inner side of the insertion frame **111***a* and extends into the corresponding insertion space **13** and a first outer side elastic finger **162** which is provided at an outer side of the insertion frame **111***a*. A plurality of snapping blocks **111***b* are formed on the outer side of the insertion frame **111***a*, the first outer elastic finger **162** is formed with a snapping hole **162***a* which correspondingly engages with the snapping block **111***b* in form of tenon and mortise joint. Each second elastic grounding member **17** has a plurality of second elastic fingers **171** which are respectively positioned on a left side and a right side of the corresponding partitioning wall **12** and extend into the corresponding insertion space **13**. The plurality of first elastic grounding members **16** and the plurality of second elastic grounding members **17** are used to correspondingly elastically contact an outer surface of the pluggable module **300** which is inserted into the insertion space **13**.

(19) In addition, a fixing structure **112***c* is formed at an outer side of each side frame portion **112** of each frame **11** of the two cages **1** and has a locking hole, and the fixing structures **112***c* of the frames **11** of the two cages **1** can be fixed relative to each other by a thread locking member (not shown), so that the two cages **1** are assembled together and are fixed relative to each other. In addition, front segments of the two side frame portions **112** of the upper cage **1***a* each are formed with a circuit board fixing structure S which extends downwardly and has a locking hole, the circuit board fixing structure S can be locked on the circuit board **200** by a thread locking member.

(20) The plurality of receptacle connectors **2** are received in the two cages **1** and are arranged side by side on the circuit board **200** along the left-right direction D**2**. Each receptacle connector **2** includes a housing **21** which is insulative and a plurality of terminals **22**. The housing **21** has two mating slots **211** which face forwardly and are arranged in the up-down direction D**1**, the two mating slots **211** respectively correspond to one of the plurality of insertion spaces **13** of the upper cage **1***a* and one of the plurality of insertion spaces **13** of the lower cage **1***b*. The plurality of terminals **22** are provided in the two mating slots **211** and tails (not shown) of the plurality of terminals **22** extend out of the housing **21** to be mechanically and electrically connected to the circuit board **200**, but the present disclosure is not limited thereto, for example, each terminal **22** of the receptacle connector **2** is not necessarily connected to the circuit board **200**, but each terminal **22** also may be connected to a cable. In addition, in the first embodiment, the connector assembly **100** further includes a rear cover **6** which is provided to and covers a rear segment of the two cages **1** and covers the plurality of receptacle connectors **2** and a plurality of rear partitioning walls **7** which are assembled to the rear cover **6** and each are positioned between two adjacent receptacle connectors **2** to partition the plurality of receptacle connectors **2** from each other. Specifically, the rear cover **6** has a shape of a substantially L-shaped plate, and the rear cover **6** is formed with a plurality of assembling holes **61** which penetrate a top portion of the rear cover **6** and a plurality of limiting grooves **62** which are formed to a front side of a rear portion of the rear cover **6**. Rear edges of the plurality of rear partitioning walls **7** snap into the plurality of limiting grooves **62** respectively, and a top edge of each rear partitioning wall **7** is formed with a plurality of assembling pieces **71** which correspondingly snap into the assembling holes **61**. In assembling, the plurality of assembling pieces **71** correspondingly pass through the assembling holes **61** and then bend so that the rear partitioning wall **7** are engaged with the rear cover **6**. In addition, the rear cover **6** is formed with two positioning posts **63** which are respectively positioned at a left side and a right side of a front end of the rear cover **6**, the two side frame portions **112** of the frame **11** positioned in the upper are formed with two positioning grooves **112***d* which respectively allow the two positioning

posts **63** to snap therein. Moreover, the rear cover **6** is further formed with a circuit board mounting structure S (see FIG. **8**) which has a locking hole, the circuit board mounting structure S can be locked to the circuit board **200** through a thread locking member **400**. When the mating portion **301** of the pluggable module **300** is inserted into the insertion space **13**, a mating circuit board **302** which is provided at a tip of the mating portion **301** of the pluggable module **300** can be inserted into the corresponding mating slot **211** to mate with the terminals **22** of the receptacle connector **2**.

(21) Referring to FIG. **1** and FIG. **5** to FIG. **8**, the first liquid cooling tray **3** is provided at a top portion of the upper cage **1***a*, a lower surface of the first liquid cooling tray **3** spans the top portion of the upper cage **1***a* and constitutes an upper wall surface of each of the plurality of insertion spaces **13** of the upper cage **1***a*. The second liquid cooling tray **4** is provided between the upper cage **1***a* and the lower cage **1***b*, an upper surface of the second liquid cooling tray **4** spans a bottom of the upper cage **1***a* and constitutes a lower wall surface of each of the plurality of insertion spaces **13** of the upper cage **1***a*, a lower surface of the second liquid cooling tray **4** spans a top portion of the lower cage **1***b* and constitutes an upper wall surface of each of the plurality of insertion spaces **13** of the lower cage **1***b*. The first liquid cooling tray **3** and the second liquid cooling tray **4** are used to circulate a cooling liquid therein, the first liquid cooling tray **3** and the second liquid cooling tray **4** each include a tray body P which allows a cooling liquid to flow therein. The cooling liquid may be for example water or other cooling liquids, the tray body P is for example a metal (for example copper or aluminum) in material and has an inlet I and an outlet O which are positioned respectively at a left side and a right side of the tray body P to allow the cooling liquid to flow in and flow out respectively, a channel R which is positioned inside the tray body P and is communicated between the inlet I and the outlet O and a plurality of heat dissipation fins F which are provided in the channel R. In the first embodiment, the tray body P is a two-piece item composed of an upper component and a lower component which can be assembled together, but the present disclosure is not limited thereto. The first liquid cooling tray **3** and the second liquid cooling tray **4** can be used together with other components of a liquid cooling heat dissipation system (not shown), so that the cooling liquid can dissipate heat via the liquid cooling heat dissipation system after absorbing heat from the first liquid cooling tray **3** and second liquid cooling tray **4** and leaving the first liquid cooling tray **3** and second liquid cooling tray **4**. The liquid cooling heat dissipation system may for example include a fluid guiding pipe, a heat dissipation row, a heat dissipation fan, a pump, a water tank and the like, the above components may be provided outside the connector assembly **100**.

(22) It is noted that, the cage **1** may also be only one in number, and at this time, the first liquid cooling tray **3** and the second liquid cooling tray **4** are respectively positioned above and below the cage **1**, the cage **1** also may be three or more in number, the first liquid cooling tray **3** and the second liquid cooling tray **4** each also may be two or more in number, in other words, the cage **1**, the first liquid cooling tray **3**, and the second liquid cooling tray **4** can be adjusted in number as desired, which is not limited to the first embodiment.

(23) Specifically, the frame **11** of each cage **1** is formed with a front assembling groove **111***c* which is positioned on an inner side surface of the front frame portion **111**, is formed forwardly and extends transversely and a plurality of rear assembling grooves **113***a* which are positioned to the rear frame portion **113** and are formed downwardly first and then forwardly. The first liquid cooling tray **3** is formed with a front assembling protruding bar **31** which is formed forwardly and snaps forwardly into the front assembling groove **111***c* of the upper cage **1***a* and a plurality of rear assembling posts **32** which correspondingly snap into the plurality of rear assembling grooves **113***a* of the upper cage **1***a* respectively, and the rear assembling post **32** has a L-shaped structure toward forwardly. The second liquid cooling tray **4** is formed with a front assembling protruding bar **41** which is formed forwardly and snaps forwardly into the front assembling groove **111***c* of the lower cage **1***b* and a plurality of rear assembling posts **42** which correspondingly snap into the rear assembling grooves **113***a* of the lower cage **1***b* respectively, the rear assembling post **42** has a L-

shaped structure toward forwardly. Specifically, a stopping block **1130** is formed at a top of a front end of each rear assembling groove **113***a*, after the rear assembling post **32** and the rear assembling post **42** which each have the L-shaped structure toward forwardly each snaps forwardly into the rear assembling groove **113***a*, each of the front assembling post **32** and the rear assembling post **42** will be stopped by the stopping block **1130** to prevent each of the front assembling post **32** and the rear assembling post **42** from being detached from the rear assembling groove **113***a*. In the first embodiment, the front assembling groove **111***c* has a main groove portion **111***d* which is formed forwardly and extends transversely and an alignment groove portion **111***e* which is formed forwardly and extends upwardly from the main groove portion **111***d*, the front assembling protruding bar **31**,**41** has a main protruding bar portion **311**, **411** which is formed forwardly and extending in the left-right direction D**2** and an alignment protruding bar portion **312**, **412** which is formed forwardly and extends upwardly from a center of the main protruding bar portion **311**, **411**, the main groove portion **111***d* corresponding cooperates with the main protruding bar portion **311**,**411**, and the alignment groove portion **111***e* correspondingly cooperates with the alignment protruding bar portion **312**,**412**, by that the alignment groove portion **111***e* and the alignment protruding bar portion **312**, **412** cooperate with each other, a position of each of the first liquid cooling tray **3** and the second liquid cooling tray **4** in the left-right direction D**2** relative to the corresponding cage **1** can be further ensured. The connector assembly **100** further includes a plurality of fixing members which are used to fix the first liquid cooling tray **3** and the upper cage **1***a* and fix the second liquid cooling tray **4** and the lower cage **1***b*. In the first embodiment, the plurality of fixing members are thread locking members **500** which correspondingly pass through and lock the first liquid cooling tray **3** and the rear frame portion **113** of the upper cage **1***a* and correspondingly pass through and lock the second liquid cooling tray **4** and the rear frame portion **113***b* of the lower cage **1***b*. When the first liquid cooling tray **3** and the second liquid cooling tray **4** are to be assembled to the upper cage **1***a* and the lower cage **1***b* respectively, the plurality of rear assembling posts **32**, **42** firstly extend downwardly into the plurality of rear assembling grooves **113***a* respectively and the front assembling protruding bar **31**, **41** aligns with the front assembling groove **111***c*, and then the first liquid cooling tray **3** or the second liquid cooling tray **4** is moved forwardly to make the front assembling protruding bar **31**, **41** snapped into the front assembling groove **111***c* and the plurality of rear assembling posts **32**, **42** snapped into the plurality of rear assembling grooves **113***a* respectively, and then the thread locking member **500** (the fixing member) passes through and locks the first liquid cooling tray **3** and the upper cage **1***a* and the thread locking member **500** (the fixing member) passes through and locks the second liquid cooling tray **4** and the lower cage **1***b*, therefore the first liquid cooling tray **3** and the second liquid cooling tray **4** are respectively assembled to the upper cage **1***a* and the lower cage **1***b*.

(24) Referring to FIG. **2**, FIG. **4**, FIG. **7** and FIG. **8**, the plurality of pressuring springs **5** are independent members, are assembled on the upper surface of the second liquid cooling tray **4** and the bottom plate portion **151** of the bottom plate **15** respectively, and are correspondingly positioned in the insertion spaces **13** respectively. Pressuring direction of the pressuring springs **5** which are assembled on the upper surface of the second liquid cooling tray **4** each are toward the first liquid cooling tray **3**, pressuring directions of the pressuring springs **5** which are assembled on the bottom plate portion **151** of the bottom plate **15** each are toward the second liquid cooling tray **4**. In the first embodiment, each pressuring spring **5** has a mounting portion **50** which is assembled to the upper surface of the second liquid cooling tray **4** or the bottom plate portion **151** of the bottom plate **15** and two spring plates **53** which extend from the mounting portion **50** and extend into the corresponding insertion space **13**. The mounting portion **50** has a plate body **51** which is assembled to the upper surface of the second liquid cooling tray **4** or the bottom plate portion **151** of the bottom plate **15** and a frame body **52** which is folded back from the plate body **51** onto an upper surface of the plate body **51**. The two spring plates **53** each rearwardly extend integrally from an inner edge of the frame body **52** and extend into the corresponding insertion space **13**. It is noted

that, in other embodiments, the spring plate **53** also may be adjusted to one or three or more in the number as desired, which is not limited to the first embodiment. In addition, in the first embodiment, a lower surface of the plate body **51** of the mounting portion **50** of the pressuring spring **5** which is assembled to the upper surface of the second liquid cooling tray **4** is provided on the second liquid cooling tray **4** by welding. The upper surface of the second liquid cooling tray **4** is formed with recessed grooves **43** which allow the plate bodies **51** of the pressuring springs **5** to be provided respectively. The bottom plate portion **151** of the bottom plate **15** is formed with windows **151***a* which correspond to the pressuring springs **5** respectively, the upper surface of the plate body **51** of the pressuring spring **5** which is assembled to the bottom plate **15** is provided to a lower surface of the bottom plate portion **151** at the window **151***a* by welding, the spring plates **53** of the pressuring spring **5** extend into the insertion space **13** via the window **151***a*, the plate body **51** of the pressuring spring **5** forms a complete plate shape which can shield the window **151***a* after assembled. However, the providing method can be adjusted as desired, for example, the plate body **51** of the pressuring spring **5** may also be provided to the second liquid cooling tray **4** or the bottom plate **15** by existing providing methods such as latching and riveting. Moreover, in a varied embodiment, the plate body **51** of the pressuring spring **5** also may be provided on the upper surface of the bottom plate portion **151** of the bottom plate **15**, so the bottom plate portion **151** of the bottom plate **15** does not need to have the window **151***a*, in addition, the upper surface of the second liquid cooling tray **4** also may not be formed with the recessed grooves **43** which allow the plate bodies **51** of the pressuring springs **5** to be provided respectively, which is not limited to the first embodiment. Furthermore, although a tip of the spring plate **53** can directly abut against the plate body **51** to be supported by the plate body **51** (see FIG. **8**) in the first embodiment, in other embodiments, the tip of the spring plate **53** also may be spaced apart from the plate body **51** by a certain distance, but only when the spring plate **53** is pressed down does the spring plate **53** abuts against the plate body **51** so as to be supported by the plate body **51**, so the present disclosure is not limited to the first embodiment. By that the first liquid cooling tray **3** and the second liquid cooling tray **4** are provided to the two cages **1** and directly constitute the wall surfaces of the insertion spaces **13** and the pressuring springs **5** correspondingly extend into the insertion spaces **13** respectively and the pressuring direction of each pressuring spring **5** is toward the first liquid cooling tray **3** or the second liquid cooling tray **4**, the pluggable module **300** (see FIG. **1**) which is inserted into the connector assembly **100** can be pushed by the pressuring spring **5** to directly contact the surface of the first liquid cooling tray **3** or the surface of the second liquid cooling tray **4**, so the first liquid cooling tray **3** or the second liquid cooling tray **4** can directly contact the heat source (the pluggable module **300**) to improve the heat dissipation efficiency, and the first liquid cooling tray **3** and the second liquid cooling tray **4** can be simplified in structure to make manufacturing and assembling of the connector assembly **100** easier.

(25) Referring to FIG. **1** and FIG. **4** to FIG. **8**, in addition, the partitioning wall **12** of the upper cage **1***a* is further assembled to the first liquid cooling tray **3** and the second liquid cooling tray **4**, and the partitioning wall **12** of the lower cage **1***b* is further assembled to the second liquid cooling tray **4** and the bottom plate portion **151** of the bottom plate **15**. A plurality of first positioning recesses **33** are formed on the lower surface of the first liquid cooling tray **3**, a plurality of second positioning recesses **44** are formed on each of the upper surface and the lower surface of the second liquid cooling tray **4**, and the bottom plate portion **151** of the bottom plate **15** is formed with a plurality of assembling holes **151***b* which penetrate the bottom plate portion **151**. Each partitioning wall **12** of the upper cage **1***a* is formed with positioning protrusions **122** which are respectively snapped into the corresponding first positioning recess **33** and the corresponding second positioning recess **44**, and each partitioning wall **12** of the lower cage **1***b* is formed with a positioning protrusion **122** which is snapped into the corresponding second positioning recess **44** and an assembling piece **123** which is snapped into the corresponding assembling hole **151***b*. It is noted that, the positioning protrusion **122** extends from the partitioning wall **12** and bends. In

addition, when the partitioning wall **12** is assembled to the bottom plate **15**, the assembling piece **123** passes through the corresponding assembling hole **151***b* and then the assembling piece **123** can be bent to make the partitioning wall **12** engaged with the bottom plate **15**.
(26) Referring to FIG. **9** to FIG. **11**, a second embodiment of the connector assembly **100** of the present disclosure differs from the first embodiment in that, in the second embodiment, the rear cover **6** is integrally formed rearwardly from a rear end of the side frame portions **112** and the rear frame portion **113** of the upper cage **1***a*. The plurality of partitioning walls **12** are longer and extend rearwardly into the rear cover **6**, the frame **11** of each cage **1** is further formed with a plurality of holding structures **115** which are positioned in a rear segment of the frame **11** and are respectively used to assemble with rear segments of the plurality of partitioning walls **12**, the holding structure **115** of the upper cage **1***a* is formed to an inner side surface of the rear cover **6**, and the holding structure **115** of the lower cage **1***b* extends rearwardly from the rear frame portion **113**. Each holding structure **115** is formed with a clamping groove **115***a* which is opened downwardly and is used to receive the partitioning wall **12**, and the clamping groove **115***a* has a plurality of lateral groove openings **115***b* facing laterally, each partitioning wall **12** is formed with a plurality of stopping pieces **124** which respectively correspond to the plurality of lateral groove openings **115***b* and obliquely extend forwardly and laterally, when the rear segment of the partitioning wall **12** is assembled to the holding structure **115** rearwardly, a top edge of the rear segment of the partitioning wall **12** is received in the clamping groove **115***a* and the plurality of stopping pieces **124** are latched with the plurality of lateral groove openings **115***b* respectively, which prevents the partitioning wall **12** from sliding relative to the frame **11**. In addition, the upper cage **1***a* further has a bottom frame strip **18** which is provided below the frame **11** and extends transversely along the left-right direction D**2**, the bottom frame strip **18** is formed with two latching portions **181** which are positioned at a left end and a right end of the bottom frame strip **18** respectively and latch with the outer side surfaces of the two side frame portions **112** of the frame **11** respectively, each latching portion **181** is formed with a latching hole **181***a* which receives a latching block **112***e* on the outer side surface of the corresponding side frame portion **112**. In addition, the partitioning wall **12** of the lower cage **1***b* is further formed with a grounding leg **125** which passes through the assembling hole **151***b* of the bottom plate portion **151** of the bottom plate **15** to insert into the circuit board **200** (see FIG. **1**).
(27) Referring to FIG. **9**, FIG. **12** and FIG. **13**, moreover, in the second embodiment, the plurality of fixing members are sliding lock members **8** which lock the first liquid cooling tray **3** and the rear frame portion **113** of the upper cage **1***a* and lock the second liquid cooling tray **4** and the rear frame portion **113** of the lower cage **1***b*. The rear frame portion **113** of each cage **1** is formed with a locking groove **113***b*, a stopping block **113***c* is formed behind a left side of the locking groove **113***b*, and two locking blocks **81** which correspond to the stopping block **113***c* are formed at a bottom of each sliding lock member **8**, and the sliding lock members **8** pass through downwardly the first liquid cooling tray **3** and the second liquid cooling tray **4** respectively and can slid in the left-right direction D**2**. When the first liquid cooling tray **3** and the second liquid cooling tray **4** are to be assembled to the two cages **1**, the plurality of rear assembling posts **32**, **42** first extend downwardly into the plurality of rear assembling grooves **113***a* respectively and the rear assembling post **31**, **32** aligns with the front assembling groove **111***c*, and then the first liquid cooling tray **3** or the second liquid cooling tray **4** is moved forwardly to make the front assembling protruding bar **31**, **41** snapped into the front assembling groove **111***c* and the plurality of rear assembling post **32**, **42** snapped into the plurality of rear assembling grooves **113***a* respectively, then the sliding lock member **8** (the fixing member) is moved toward the left to make the locking block **81** of the sliding lock member **8** lock the stopping block **113***c* at the locking groove **113***b* from below, so that the first liquid cooling tray **3** and the second liquid cooling tray **4** are assembled to the two cages **1** respectively.
(28) Referring to FIG. **9** to FIG. **12**, in addition, in the second embodiment, the mounting portion

54 of each pressuring spring **5** is positioned at a front end of each pressuring spring **5**, and the spring plate **55** integrally extends rearwardly from the mounting portion **54** and extends into the corresponding insertion space **13**. In addition, specifically, the mounting portion **54** of the pressuring spring **5** is assembled to the upper surface of the second liquid cooling tray **4** or the upper surface of the bottom plate portion **151** of the bottom plate **15** by welding, however, in other embodiments, the mounting portion **54** of the pressuring spring **5** also can be assembled by existing providing methods such as latching, riveting and the like, but the present disclosure is not limited thereto. Providing grooves **45** are formed downwardly at a front end of the upper surface of the second liquid cooling tray **4** and receive the mounting portions **54** of the pressuring springs **5** respectively, so as to strengthen holding force of the second liquid cooling tray **4** with respect to the pressuring springs **5**.
(29) In conclusion, in the present disclosure, by that the first liquid cooling tray **3** and the second liquid cooling tray **4** are provided to the two cages **1** and directly constitute the wall surfaces of the insertion spaces **13** and the pressuring springs **5** correspond to the insertion spaces **13** respectively and the pressuring direction of each pressuring spring **5** is toward the first liquid cooling tray **3** or the second liquid cooling tray **4**, the pluggable module **300** which is inserted into the connector assembly **100** can be pushed by the pressuring spring **5** to directly contact the surface of the first liquid cooling tray **3** or the surface of the second liquid cooling tray **4**, so the first liquid cooling tray **3** or the second liquid cooling tray **4** can directly contact the heat source (the pluggable module **300**) to improve the heat dissipation efficiency, and the first liquid cooling tray **3** and the second liquid cooling tray **4** can be simplified in structure to make manufacturing and assembling of the connector assembly **100** easier.
(30) However, the above description is only for the embodiments of the present disclosure, and it is not intended to limit the implementing scope of the present disclosure, and the simple equivalent changes and modifications made according to the claims and the contents of the specification are still included in the scope of the present disclosure.

Claims

1. A connector assembly, comprising: a cage comprising a frame and a plurality of partitioning walls, the frame and the plurality of partitioning walls defining an insertion space; a first liquid cooling tray provided to a top of the cage, a lower surface of the first liquid cooling tray constituting an upper wall surface of the insertion space; a second liquid cooling tray provided to a bottom of the cage, an upper surface of the second liquid cooling tray constituting a lower wall surface of the insertion space; and a pressuring spring within the insertion space.
2. The connector assembly of claim 1, wherein the pressuring spring is provided on the upper surface of the second liquid cooling tray within the insertion space.
3. The connector assembly of claim 1, wherein the pressuring spring comprises a plurality of spring plates extending into the insertion space from the upper surface of the second liquid cooling tray.
4. The connector assembly of claim 1, wherein the pressuring spring comprises a mounting portion assembled to the upper surface of the second liquid cooling tray and a spring plate extending from the mounting portion into the insertion space.
5. The connector assembly of claim 4, wherein: the mounting portion of the pressuring spring comprises a plate body assembled to the upper surface of the second liquid cooling tray and a frame body folded back from the plate body onto an upper surface of the plate body; and the spring plate extends from an inner edge of the frame body into the insertion space.
6. The connector assembly of claim 4, wherein: the mounting portion of the pressuring spring is positioned at a front end of the pressuring spring; and the spring plate extends from the mounting portion into the insertion space.
7. The connector assembly of claim 1, wherein the plurality of partitioning walls are assembled to

the first liquid cooling tray and the second liquid cooling tray.
8. The connector assembly of claim 1, wherein: a first positioning recess is formed on the lower surface of the first liquid cooling tray; a second positioning recess is formed on the upper surface of the second liquid cooling tray; and a partitioning wall among the plurality of partitioning walls comprises positioning protrusions which are snapped into the first positioning recess and the second positioning recess.
9. The connector assembly of claim 1, wherein: the plurality of partitioning walls define a plurality of insertion spaces; the lower surface of the first liquid cooling tray constitutes an upper wall surface of the plurality of insertion spaces; and the upper surface of the second liquid cooling tray constitutes a lower wall surface of the plurality of insertion spaces.
10. The connector assembly of claim 9, wherein: the insertion space comprises a first insertion space among the plurality of insertion spaces; and the connector assembly further comprises a second pressuring spring within a second insertion space among the plurality of insertion spaces.
11. A connector assembly, comprising: a cage comprising an upper cage and a lower cage, the cage comprising a frame and a plurality of partitioning walls, the frame and the plurality of partitioning walls defining an upper insertion space in the upper cage and a lower insertion space in the lower cage; a first liquid cooling tray provided to a top of the upper cage, a lower surface of the first liquid cooling tray constituting an upper wall surface of the upper insertion space; a second liquid cooling tray provided between the upper cage and the lower cage, an upper surface of the second liquid cooling tray constituting a lower wall surface of the upper insertion space, a lower surface of the second liquid cooling tray constituting an upper wall surface of the lower insertion space; a first pressuring spring within the upper insertion space; and a second pressuring spring within the lower insertion space.
12. The connector assembly of claim 11, wherein: the first pressuring spring is provided on the upper surface of the second liquid cooling tray within the upper insertion space; and the second pressuring spring is provided on the lower surface of the second liquid cooling tray within the lower insertion space.
13. The connector assembly of claim 11, wherein: the first pressuring spring comprises a first plurality of spring plates extending into the upper insertion space from the upper surface of the second liquid cooling tray; and the second pressuring spring comprises a second plurality of spring plates extending into the lower insertion space from the lower surface of the second liquid cooling tray.
14. The connector assembly of claim 11, wherein: the first pressuring spring comprises a first mounting portion assembled to the upper surface of the second liquid cooling tray and a first spring plate extending from the first mounting portion into the upper insertion space; and the second pressuring spring comprises a second mounting portion assembled to the lower surface of the second liquid cooling tray and a second spring plate extending from the second mounting portion into the lower insertion space.
15. The connector assembly of claim 11, wherein the plurality of partitioning walls are assembled to the first liquid cooling tray and the second liquid cooling tray.
16. The connector assembly of claim 11, wherein: a first positioning recess is formed on the lower surface of the first liquid cooling tray; a second positioning recess is formed on the upper surface of the second liquid cooling tray; and a partitioning wall among the plurality of partitioning walls comprises positioning protrusions which are snapped into the first positioning recess and the second positioning recess.
17. The connector assembly of claim 11, wherein: the upper insertion space comprises plurality of upper insertion spaces; the lower surface of the first liquid cooling tray constitutes an upper wall surface of the plurality of upper insertion spaces; and the upper surface of the second liquid cooling tray constitutes a lower wall surface of the plurality of upper insertion spaces.
18. A connector assembly, comprising: a cage comprising a frame and a plurality of partitioning

walls, the frame and the plurality of partitioning walls defining an insertion space; a liquid cooling tray provided to a top of the cage, a lower surface of the liquid cooling tray constituting an upper wall surface of the insertion space; and a pressuring spring within the insertion space.

19. The connector assembly of claim 18, further comprising a second liquid cooling tray provided to a bottom of the cage, an upper surface of the second liquid cooling tray constituting a lower wall surface of the insertion space.

20. The connector assembly of claim 19, wherein the pressuring spring is provided on the upper surface of the second liquid cooling tray within the insertion space.